and denote by N the cone of space X, consisting of the single point 0. It is clear from $(57)_i$ that the condition indicated in Problem 2 implies that the operator S is ε-invariant for the constraints $\|u\|\leqslant l$, $Tu\in N$. Using Theorem 4, we obtain the following result: the condition indicated in Problem 2 is satisfied if and only if a functional $\bar{x}^*\in X^*$ exists with the property

$$\sum_{i=0}^{m-1} |(\bar{x}^*, A_1^i b_1)-(x^*, A_2^i b_2)|\leqslant \varepsilon/l.$$

Translated by D. E. Brown

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EXPANSION WITH RESPECT TO A PARAMETER OF THE EXTREMAL VALUES OF GAME PROBLEMS*

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THE EXTREMAL value of a parametric problem of mathematical programming is considered as a function of a parameter. The directional derivatives of different orders of this function are investigated.

Consider the function

$$\varphi(x)=\max_{y\in\Omega(x)} f(x,y),$$

where $\Omega(x)$ are closed sets of E_y, dependent on the parameter $x\in E_x$; E_x, E_y are finite-dimensional Euclidean spaces. We are interested in the first directional derivative $\partial\varphi(x_0)/\partial g$. This problem was discussed in [1–10]. If $\partial\varphi(x_0)/\partial g$ exists and is finite, then the limit

$$\frac{\partial^2\varphi(x_0)}{\partial g^2}=\lim_{t\to+0}\frac{2}{t^2}\left[\varphi(x_0+tg)-\varphi(x_0)-\frac{\partial\varphi(x_0)}{\partial g}t\right] \tag{0.1}$$

will be called the second directional derivative. If the limit (0.1) exists and is finite, then

$$\varphi(x_0+tg)=\varphi(x_0)+\frac{\partial\varphi(x_0)}{\partial g}t+\frac{1}{2}\frac{\partial^2\varphi(x_0)}{\partial g^2}t^2+o(t^2).$$

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The problem of the existence and determination of the second derivative was touched on in [6–8]. The higher-order derivatives are similarly defined.

It will be assumed henceforth that the point-set mapping Ω is continuous at $x = x_0$ (see [11]), that the sets $\Omega(x_0+tg)$ are contained in a bounded set D for $0 \leqslant t < t_1$, and that the function $f(x, y)$ is continuous. These conditions ensure that the function $\varphi(x_0+tg)$ is continuous for $t = +0$.

1. First directional derivative

1. We fix a point x_0 and a direction g. Let $y \in \Omega(x_0)$. As in [6–7], we introduce the set

$$\gamma(y) = \{v \mid \exists t_0 > 0\colon\ y + tv \in \Omega(x_0+tg) \ \text{ for } \ 0 \leqslant t < t_0\}.$$

We denote the closure of $\gamma(y)$ by $\Gamma(y)$ and call it the set of admissible directions. The set $\Gamma(y)$ is not in general a cone and may prove to be empty.

Definition [6–7]. We shall say that the mapping Ω admits of first-order approximation at the point $\bar{y} \in \Omega(x_0)$ if, given any sequence $\{y_k\}$ such that

$$y_k \in \Omega(x_0+t_k g), \quad t_k \to +0, \quad y_k \to \bar{y}, \tag{1.1}$$

we have

$$y_k = \bar{y} + t_k v_k + o(t_k), \tag{1.2}$$

where $v_k \in \Gamma(\bar{y})$, $t_k v_k \to 0$.

We introduce the set

$$R(x_0) = \{y \in \Omega(x_0) \mid \varphi(x_0) = f(x_0, y)\}.$$

The following was proved in [7]:

Theorem 1

Let the mapping Ω admit of first-order approximation at every point $\bar{y} \in R(x_0)$, let the function $f(x, y)$ be continuously differentiable, and given any sequence $\{y_k\}$ of the form (1.1), (1.2), where $\bar{y} \in R(x_0)$, let

$$f(x_0+t_k g, y_k) - f(x_0, \bar{y}) \leqslant t_k[(f_x(x_0, \bar{y}), g) + (f_y(x_0, \bar{y}), v_k)] + o(t_k). \tag{1.3}$$

Then $\partial\varphi(x_0)/\partial g$ exists and

$$\frac{\partial\varphi(x_0)}{\partial g} = \max_{y \in R(x_0)} \sup_{v \in \Gamma(y)} [(f_x(x_0, y), g) + (f_y(x_0, y), v)], \tag{1.4}$$

where the maximum with respect to $y \in R(x_0)$ is reached.

Notes. 1. Condition (1.3) is satisfied e.g., if the function $f(x_0, y)$ is concave in y in the neighbourhood of every point $\bar{y} \in R(x_0)$.

2. An expression similar to (1.4) was also obtained in [1], under the assumption that the sets $\Omega(x)$ are specified by convex inequalities, and that $f(x, y)$ is concave in y for every x.

2. Now let the set $\Omega(x)$ have the form

$$\Omega(x)=M\cap\{y\,|\,h_i(x, y)\leqslant 0,\quad i=1, 2, \ldots, m\},$$

where M is a convex closed set independent of x, and the functions h_i are continuously differentiable.

Assumption 1. For every point $\bar{y}\in R(x_0)$, for which $Q(\bar{y})=\{i\,|\,h_i(x_0, \bar{y})=0\}\neq\varnothing$, a v exists (dependent on $\bar{y}$) such that

$$v\in K_M(\bar{y}),\qquad (h_{iy}(x_0, \bar{y}), v)>0\quad \forall i\in Q(\bar{y}), \tag{1.5}$$

where $K_M(\bar{y})$ is the cone of admissible directions for M at the point $\bar{y}$. The existence of such a v is equivalent to the condition

$$K_M^*(\bar{y})\cap\mathrm{co}\{h_{iy}(x_0, \bar{y})\,|\,i\in Q(\bar{y})\}=\varnothing,$$

i.e., to the fact that $\bar{y}$ is not a stationary point on the set M of the function

$$h(y)=\max_{i=1,2,\ldots,m} h_i(x_0, y).$$

Here, $K_M^*(\bar{y})$ is the cone conjugate to the cone $K_M(\bar{y})$.

With this assumption, it is easy to prove the following proposition, analogous to Lemma 1 of [7].

Lemma 1

Given any $\bar{y}\in R(x_0)$ the set $\Gamma(\bar{y})$ is non-empty and has the form

$$\Gamma(\bar{y})=K_M(\bar{y})\cap\{v\,|\,(h_{ix}(x_0, \bar{y}), g)+(h_{iy}(x_0, \bar{y}), v)\leqslant 0,\quad i\in Q(\bar{y})\}.$$

Assumption 2. Given any $\bar{y}\in R(x_0)$ and any sequence (1.1), we have

$$h_i(x_0, y_k)-h_i(x_0, \bar{y})\geqslant(h_{iy}(x_0, \bar{y}), y_k-\bar{y})$$

for $i\in Q(\bar{y})$ and sufficiently large k.

This condition is satisfied e.g., if the functions $h_i(x_0, y)$, $i\in Q(\bar{y})$, are convex in y in the neighbourhood of the point $\bar{y}$.

Lemma 2

If assumptions 1 and 2 hold, the mapping Ω admits of first-order approximation at every point $\bar{y}\in R(x_0)$.

Proof. Let $\bar{y}\in R(x_0)$ and let (1.1) hold. We write y_k as $y_k=\bar{y}+t_k v_k'$. Clearly, $v_k'\in K_M(\bar{y})$. For $i\in Q(\bar{y})$ we have

$$0\geqslant h_i(x_0+t_k g, y_k)=h_i(x_0+t_k g, y_k)-h_i(x_0, y_k)$$
$$+h_i(x_0, y_k)-h_i(x_0, \bar{y})\geqslant t_k(h_{ix}(x_0, \bar{y}), g)+o(t_k)+t_k(h_{iy}(x_0, \bar{y}), v_k').$$

Hence

$$(h_{ix}(x_0, \bar{y}), g)+(h_{iy}(x_0, \bar{y}), v_k')\leqslant\varepsilon_k,\qquad \varepsilon_k\to 0.$$

From (1.5), a v_0 exists such that

$$v_0 \in K_M(\bar{y}), \qquad (h_{iy}(x_0, \bar{y}), v_0) < -1, \qquad i \in Q(\bar{y}).$$

We now put $v_k = v_k' + |\varepsilon_k| v_0$. Then (1.2) will hold, while $v_k \in \Gamma(\bar{y})$.

Theorem 2

If assumptions 1 and 2 are satisfied, the function f is continuously differentiable, and given any sequence of the type (1.1), (1.2), where $\bar{y} \in R(x_0)$, the relation (1.3) is satisfied, then there exists $\partial\varphi(x_0)/\partial g$ and

$$\frac{\partial\varphi(x_0)}{\partial g} = \max_{y \in R(x_0)} \min_{\lambda \in \Lambda(y)} \left(f_x(x_0, y) - \sum_{i=1}^{m} \lambda_i h_{ix}(x_0, y), g \right),$$

where $\Lambda(y)$ is the set of Lagrange multipliers:

$$\Lambda(y) = \Big\{ \lambda \,|\, -f_y(x_0, y) + \sum_{i=1}^{m} \lambda_i h_{iy}(x_0, y) \in K_M^*(y), \lambda_i \geqslant 0, \lambda_i h_i(x_0, y) = 0, i = 1, 2, \ldots, m \Big\}.$$

Proof. Since the assumptions of Theorem 1 are satisfied, we have (1.4). To find

$$\sup_{v \in \Gamma(\bar{y})} (f_y(x_0, \bar{y}), v), \tag{1.6}$$

where $\Gamma(\bar{y})$ is described in Lemma 1, we have to solve a problem of linear programming. The dual problem amounts to finding

$$\inf_{\lambda \in \Lambda(\bar{y})} \left[-\sum_{i=1}^{m} \lambda_i (h_{ix}(x_0, \bar{y}), g) \right] \tag{1.7}$$

(see e.g., [2], pp. 9–11). The set $\Lambda(\bar{y})$ is non-empty for $\bar{y} \in R(x_0)$. For, by the necessary condition for a maximum, with $\bar{y} \in R(x_0)$ there exist numbers λ_0, λ_i, not all zero, such that

$$\begin{gathered} -\lambda_0 f_y(x_0, \bar{y}) + \sum_{i=1}^{m} \lambda_i h_{iy}(x_0, \bar{y}) \in K_M^*(\bar{y}), \\ \lambda_0 \geqslant 0, \quad \lambda_i \geqslant 0, \quad \lambda_i h_i(x_0, \bar{y}) = 0, \quad i = 1, 2, \ldots, m \end{gathered} \tag{1.8}$$

(see [12], pp. 67–70). If $\lambda_0 = 0$, then, from (1.5),

$$v \in K_M(\bar{y}), \qquad \left(\sum_{i=1}^{m} \lambda_i h_{iy}(x_0, \bar{y}), v \right) < 0,$$

which contradicts (1.8) with $\lambda_0 = 0$. Hence $\lambda_0 > 0$, and we can take $\lambda_0 = 1$, and this means that $\Lambda(\bar{y}) \neq \varnothing$. The condition (1.5) implies that the problem (1.6) satisfies Slater's condition. Hence the quantities (1.6) and (1.7) are equal and finite, and the infimum (1.7) is reached (see [2], p. 27). The theorem is proved.

Note. All the results of Paragraphs 1 and 2 extend virtually word for word to the infinite-dimensional case.

3. Let the convex set $X \subset E_x$. Let us find the direction of fastest descent of the function $\varphi(x)$ in X at the point $x_0 \in X$ and obtain the necessary condition for $\varphi(x)$ to have a minimum in X.

Let $K(x_0)$ be the cone of admissible directions for X at the point x_0, and $K_1(x_0) = \{g \in \overline{K(x_0)} \mid \|g\| \leq 1\}$, where the bar denotes closure, and $\|g\|$ is the Euclidean norm.

Assume that, given any $g \in K_1(x_0)$ the conditions of Theorem 2 hold. Then, for all $g \in K_1(x_0)$

$$\frac{\partial \varphi(x_0)}{\partial g} = \max_{y \in R(x_0)} \min_{\lambda \in \Lambda(y)} (L_x(x_0, y, \lambda), g),$$

$$L(x, y, \lambda) = f(x, y) - \sum_{i=1}^{m} \lambda_i h_i(x, y).$$

We shall consider functions $\lambda(\cdot)$, specified on $R(x_0)$ and such that $\lambda(y) \in \Lambda(y)$ for all $y \in R(x_0)$. It is easily shown that

$$\frac{\partial \varphi(x_0)}{\partial g} = \min_{\lambda(\cdot)} \sup_{y \in R(x_0)} (L_x(x_0, y, \lambda(y)), g). \tag{1.9}$$

The minimum with respect to the functions $\lambda(\cdot)$ is reached, since the minimum with respect to $\lambda \in \Lambda(y)$ is reached for any $y \in R(x_0)$. With every function $\lambda(\cdot)$ we link a set

$$P = \overline{\mathrm{co}}\{L_x(x_0, y, \lambda(y)) \mid y \in R(x_0)\}.$$

It can be shown that, by virtue of (1.5), the sets $\Lambda(y)$, $y \in R(x_0)$, are bounded in aggregate. Hence the sets P are convex compacta and are also bounded in aggregate.

We can now write (1.9) in the form

$$\partial \varphi(x_0)/\partial g = \min_{P} \max_{v \in P} (v, g). \tag{1.10}$$

Hence $\partial \varphi(x_0)/\partial g$ regarded as a function of g, satisfies a Lipschitz condition. Hence $\min_{g \in K_1(x_0)} \frac{\partial \varphi(x_0)}{\partial g}$ is reached, i.e., there exist directions of fastest descent. From (1.10),

$$\min_{g \in K_1(x_0)} \frac{\partial \varphi(x_0)}{\partial g} = \min_{g \in K_1(x_0)} \min_{P} \max_{v \in P} (v, g) = \min_{P} \min_{g \in K_1(x_0)} \max_{v \in P} (v, g).$$

We introduce the distance $\rho(P)$ from the conjugate cone $K^*(x_0)$ to P:

$$\rho(P) = \min_{w \in K^*(x_0)} \min_{v \in P} \|w - v\| = \|w(P) - v(P)\|,$$

where $w(P) \in K^*(x_0)$, $v(P) \in P$. Then

$$\min_{g \in K_1(x_0)} \max_{v \in P} (v, g) = -\rho(P),$$

and if $\rho(P) > 0$, then the minimum is reached at the unique point

$$g(P) = \frac{w(P) - v(P)}{\|w(P) - v(P)\|}$$

(see [13], p. 152).

From the above discussion we easily arrive at:

Theorem 3

$$\min_{g\in K_1(x_0)} \frac{\partial\varphi(x_0)}{\partial g} = -\max_P \rho(P). \tag{1.11}$$

If the maximum is positive, then the set

$$\{g(P_0) \mid \rho(P_0) = \max_P \rho(P)\}$$

is the set of directions of fastest descent of the function $\varphi(x)$ in X at the point x_0.

In short, to find the directions of fastest descent, we have to seek the compacta P_0, furthest away from $K^*(x_0)$, then construct the vectors $g(P_0)$.

If $\varphi(x)$ reaches its minimum in X at the point x_0, the left- and right-hand sides of (1.11) vanish. It immediately follows that the necessary condition for a minimum is

$$K^*(x_0)\cap P\neq\varnothing \quad \text{for all} \quad P. \tag{1.12}$$

4. Now assume that the set X is not convex. To obtain the necessary condition for a minimum, we use the cone of directions, admissible in the wide sense (see [4]).

Let $x_0\in X$. We call a vector g a direction admissible in the wide sense if, given any $\varepsilon>0$ a number t_ε and a vector g_ε exist such that

$$x_0+t_\varepsilon g_\varepsilon\in X, \qquad 0<t_\varepsilon<\varepsilon, \qquad \|g_\varepsilon-g\|<\varepsilon. \tag{1.13}$$

A cone of directions, admissible in the wide sense, will be denoted by $M(x_0)$. Notice that $M(x_0)$ is a closed cone.

We can write the cone $M(x_0)$ as the union of a finite or infinite number of convex closed cones $M_\alpha(x_0)$:

$$M(x_0) = \bigcup_\alpha M_\alpha(x_0).$$

For example, we can take as $M_\alpha(x_0)$ the rays which make up the cone $M(x_0)$. However, the minimal representation of $M(x_0)$ is naturally of the greatest interest. We have:

Proposition. Let the function $\varphi(x)$ satisfy a Lipschitz condition in the neighbourhood of the point x_0. Then, in order for $\varphi(x)$ to reach a minimum on X at the point x_0, we must have

$$\inf_{g\in M_\alpha(x_0)} \frac{\partial\varphi(x_0)}{\partial g} = 0 \qquad \text{for all} \quad \alpha. \tag{1.14}$$

Condition (1.14) is equivalent to

$$M_\alpha^*(x_0)\cap P\neq\varnothing \quad \text{for all} \quad P \text{ and } \alpha. \tag{1.15}$$

For, if (1.14) is infringed, there must exist, for some α, a $g\in M_\alpha(x_0)$, for which $\partial\varphi(x_0)/\partial g<0$. For this g, there exist t_ε and g_ε, satisfying (1.13). Then, in view of the Lipschitz condition, for small ε

$$\varphi(x_0+t_\varepsilon g_\varepsilon)\leqslant\varphi(x_0+t_\varepsilon g)+Lt_\varepsilon\|g_\varepsilon-g\|<\varphi(x_0).$$

This contradiction proves (1.14). The rest of the proof of the equivalence of (1.14) and (1.15) is obtained, as above, by using (1.10).

Note. The idea of separating convex cones $M_\alpha(x_0)$ from the cone of directions, admissible in the wide sense, is exactly similar to Boltyanskii's idea [15], of considering a number of "tents", which are convex cones. These "tents" were used in [15] to obtain a necessary condition for a smooth function to have a minimum on a set of quite general type.

2. Second directional derivatives

Let $y \in \Omega(x_0)$, $v \in \Gamma(y)$. We introduce the set

$$\gamma_2(y, v) = \{w \mid \exists t_c > 0\colon\ y + tv + t^2 w \in \Omega(x_0 + tg) \quad \exists t \in [0, t_0)\}.$$

We denote the closure of $\gamma_2(y, v)$ by $\Gamma_2(y, v)$, and call it a set of admissible directions of the second order. When $\Omega(x)$ is independent of x, we get a set similar to that discussed in [16].

Definition [6]. We shall say that the mapping Ω admits of second-order approximation at the point $\bar{y} \in \Omega(x_0)$, if, given any sequence $\{y_k\}$ such that

$$y_k \in \Omega(x_0 + t_k g), \qquad t_k \to +0, \qquad y_k = \bar{y} + t_k \bar{v} + o(t_k), \qquad \bar{v} \in \Gamma(\bar{y}), \tag{2.1}$$

we have

$$y_k = \bar{y} + t_k \bar{v} + t_k^2 w_k + o(t_k^2), \qquad w_k \in \Gamma_2(\bar{y}, \bar{v}), \qquad t_k w_k \to 0. \tag{2.2}$$

Now let the sets $\Omega(x)$ be specified by the inequalities

$$\Omega(x) = \{y \mid h_i(x, y) \leqslant 0,\ i = 1, 2, \ldots, m\},$$

where the functions $h_i(x, y)$ are twice continuously differentiable.

Assumption 3. For every point $\bar{v} \in R(x_0)$, for which $Q(\bar{y}) \neq \varnothing$, a $v = v(\bar{y})$ exists such that

$$(h_{iy}(x_0, \bar{y}), v) < 0, \quad i \in Q(\bar{y}) = \{i \mid h_i(x_0, y) = 0\}. \tag{2.3}$$

The following two simple lemmas hold under this assumption.

Lemma 3

Given any $\bar{y} \in R(x_0)$, $v \in \Gamma(\bar{y})$ the set $\Gamma_2(\bar{y}, v)$ is non-empty and has the form

$$\Gamma_2(\bar{y}, v) = \{w \mid 2(h_{iy}, w) + (h_{iyy} v, v) + 2(h_{ixy} v, g) + (h_{ixx} g, g) \leqslant 0,\ i \in Q_2(\bar{y}, v)\},$$

where

$$Q_2(\bar{y}, v) = \{i \in Q(\bar{y}) \mid (h_{iy}, v) + (h_{ix}, g) = 0\},$$

and all the derivatives are taken at the point $[x_0, \bar{y}]$.

Lemma 4

The mapping Ω admits of second-order approximation at every point $\bar{y}\in R(x_0)$.

The proof is similar to the proof of Lemma 2.

If $\partial\varphi(x_0)/\partial g$ exists and (1.4) holds, we shall denote by T the sets of pairs $[y, v]$, on which the extremum in (1.4) is reached:

$$T=\left\{[y,v]\;\middle|\;\frac{\partial\varphi(x_0)}{\partial g}=(f_x(x_0,y),g)+(f_y(x_0,y),v),\;\; y\in R(x_0),\;\; v\in\Gamma(y)\right\}.$$

Assumption 4. Given any $\bar{y}\in R(x_0)$ the functions $h_i(x_0+tg,\; y)$, $i\in Q(\bar{y})$, are convex in y in some neighbourhood of the point $\bar{y}$, whatever the fixed t, $0\leqslant t<t_1$.

Theorem 4

Let assumptions 3 and 4 be satisfied, let the function f be twice continuously differentiable, and for any $\bar{y}\in R(x_0)$ let the matrix $f_{yy}(x,\bar{y})$ be negative definite. Then there exists the second derivative $\partial^2\varphi(x_0)/\partial g^2$, where

$$\frac{\partial^2\varphi(x_0)}{\partial g^2}=\max_{[y,v]\in T}\;\max_{w\in\Gamma_2(y,v)}\;[2(f_y,w)+(f_{yy}v,v)+2(f_{xy}v,g)+(f_{xx}g,g)],\tag{2.4}$$

and all the derivatives are taken at the point $[x_0, y]$.

Proof. We introduce the notation

$$\Phi(y,v,w)=2(f_y,w)+(f_{yy}v,v)+2(f_{xy}v,g)+(f_{xx}g,g).$$

Notice that the set T is not empty: the maximum with respect to $y\in R(x_0)$ in (1.4) is reached, by Theorem 1. Discovery of the supremum with respect to $v\in\Gamma(y)$ is a problem of the type (3.1), (3.2). Hence the maximum with respect to $v\in\Gamma(y)$ is reached for any $y\in R(x_0)$.

We fix $[y, v]\in T$ and $w\in\gamma_2(y, v)$. Then, $y+tv+t^2w\in\Omega(x_0+tg)$ for small t. Hence

$$\begin{aligned}&\varphi(x_0+tg)\geqslant f(x_0+tg,y+tv+t^2w)=\varphi(x_0)+t\frac{\partial\varphi(x_0)}{\partial g}\\&+\frac{1}{2}t^2\Phi(y,v,w)+o(t^2),\\&\underline{A}=\varliminf_{t\to+0}\frac{2}{t^2}\left[\varphi(x_0+tg)-\varphi(x_0)-t\frac{\partial\varphi(x_0)}{\partial g}\right]\geqslant\Phi(y,v,w).\end{aligned}\tag{2.5}$$

Since $[y, v]\in T$ is arbitrary and $w\in\gamma_2(x, v)$, we find that $\underline{A}\geqslant A$, where A denotes the right-hand side of (2.4).

Let us show that $\bar{A}\leqslant A$. We choose $\{y_k\}$ and $\{t_k\}$ in such a way that

$$\frac{2}{t^2}\left[\varphi(x_0+t_kg)-\varphi(x_0)-t_k\frac{\partial\varphi(x_0)}{\partial g}\right]\to\bar{A},\quad y_k\in R(x_0+t_kg),\quad y_k\to\bar{y}.$$

Clearly, $\bar{y}\in R(x_0)$. We write y_k in the form $y_k=\bar{y}+t_kv_k'$, $t_kv_k'\to 0$ For $i\in Q(\bar{y})$

$$0\geqslant h_i(x_0+t_kg,y_k)-h_i(x_0,\bar{y})=t_k[(h_{ix}(x_0,\;\bar{y}),\;g)+(h_{iy}(x_0,\bar{y}),\;v_k')]$$

$$+\frac{1}{2}t_k^2[(h_{iyy}v_k',v_k')+2(h_{ixy}v_k',g)+(h_{ixx}g,g)],$$

where the second derivatives are taken at the point

$$\theta_{ik}=[x_0+\tau_{ik}g,\ \bar{y}+\tau_{ik}v_k'],\qquad 0<\tau_{ik}<t_k. \tag{2.6}$$

By assumption 4, $(h_{iyy}v_k',\ v_k')\geqslant 0$. Hence

$$\begin{gathered}(h_{ix},g)+(h_{iy},v_k')\leqslant\varepsilon_k=t_k\max_i\left[-(h_{ixy}v_k',g)-\frac{1}{2}(h_{ixx}g,g)\right],\\ |\varepsilon_k|\leqslant t_k[C_1\|v_k'\|+C_2].\end{gathered} \tag{2.7}$$

By (2,3), a v_0 exists such that

$$(h_{iy}(x_0,\bar{y}),\ v_0)\leqslant -1,\qquad i\in Q(\bar{y}).$$

We set $v_k=v_k'+|\varepsilon_k|v_0$. By Lemma 1, $v_k\in\Gamma(\bar{y})$. Further,

$$\begin{gathered}\varphi(x_0+t_kg)=f(x_0+t_kg,\ y_k)=f(x_0,\bar{y})+t_k[(f_x,g)+(f_y,v_k)]\\ -t_k|\varepsilon_k|(f_y,\ v_0)+{}^1/_2t_k^2[(f_{yy}v_k'',\ v_k')+2(f_{xy}v_k',\ g)+(f_{xx}g,\ g)],\end{gathered}$$

where the second derivatives are taken at a point of the type (2.6). By (1.4), $(f_x,g)+(f_y,v_k)\leqslant \partial\varphi(x_0)/\partial g$. For large k,

$$(f_{yy}v_k',v_k')\leqslant -\frac{m}{2}\|v_k'\|^2,\qquad m>0. \tag{2.8}$$

Using (2.7) and (2.8), we get

$$\varphi(x_0+t_kg)\leqslant\varphi(x_0)+t_k\frac{\partial\varphi(x_0)}{\partial g}+\frac{1}{2}t_k^2\left[-\frac{m}{2}\|v_k'\|^2+C_3\|v_k'\|+C_4\right]. \tag{2.9}$$

Comparing (2.5) and (2.9), we find that

$$\Phi(y,v,w)+o(1)\leqslant -\frac{m}{2}\|v_k'\|^2+C_3\|v_k'\|+C_4.$$

The sequence $\{v_k'\}$ is thus bounded.

We can assume that $v_k'\to\bar{v}$. Hence also, $v_k\to\bar{v}$. But $v_k\in\Gamma(\bar{y})$, so that also $\bar{v}\in\Gamma(\bar{y})$. Further, we have

$$\varphi(x_0+t_kg)=\varphi(x_0)+t_k\frac{\partial\varphi(x_0)}{\partial g}+o(t_k),$$

$$\varphi(x_0+t_kg)=f(x_0+t_kg,\ y_k)=\varphi(x_0)+t_k[(f_y,v_k)+(f_x,g)]+o(t_k).$$

Hence

$$\lim_{k\to\infty}[(f_x,g)+(f_y,v_k)]=(f_x,g)+(f_y,\bar{y})=\frac{\partial\varphi(x_0)}{\partial g},$$

i.e., $[\bar{y},\ \bar{v}]\in T$. We now write y_k in the form $y_k=\bar{y}+t_k\bar{v}+o(t_k)$. By Lemma 4, we have

$$y_k=\bar{y}+t_k\bar{v}+t_k^2w_k+o(t_k^2),\qquad w_k\in\Gamma_2(\bar{y},\bar{v}),\qquad t_kw_k\to 0.$$

Hence

$$\varphi(x_0+t_kg)=\varphi(x_0)+t_k\frac{\partial\varphi(x_0)}{\partial g}+{}^1/_2t_k^2\Phi(\bar{y},\bar{v},w_k)+o(t_k^2),$$

$$\bar{A}=\lim\frac{2}{t^2}\left[\varphi(x_0+t_kg)-\varphi(x_0)-t_k\frac{\partial\varphi(x_0)}{\partial g}\right]\leqslant\sup_{w\in\Gamma_2(\bar{y},\bar{v})}\Phi(\bar{y},\bar{v},w).$$

The theorem is proved.

3. Using the Lagrange multipliers

Let the assumptions of Theorem 4 hold. Then $\partial\varphi(x_0)/\partial g$ exists and (1.4) holds. We fix $y\in R(x_0)$ and consider the problem of linear programming

$$\max_{v\in\Gamma(y)} (f_y(x_0, y), v), \tag{3.1}$$

$$\Gamma(y)=\{v \mid (h_{ix}(x_0, y), g)+(h_{iy}(x_0, y), v)\leqslant 0,\ i\in Q(y)\} \tag{3.2}$$

and the dual problem

$$\min_{\lambda\in\Lambda(y)} \left[-\sum_{i=1}^{m} \lambda_i (h_{ix}(x_0, y), g) \right], \tag{3.3}$$

$$\Lambda(y)=\left\{\lambda \,\middle|\, \sum_{i=1}^{m} \lambda_i h_{iy}(x_0, y)=f_y(x_0, y),\ \lambda_i\geqslant 0,\ \lambda_i h_i(x_0, y)=0\right\}. \tag{3.4}$$

The sets $\Gamma(y)$ and $\Lambda(y)$ are not empty, so that both the problems have solutions.

Let $V_2(y)$ be the set of solutions of problem (3.1), (3.2), and $\Lambda_2(y)$ the set of solutions of problem (3.3), (3.4), while $R_2(x_0)$ is the set of the $y\in R(x_0)$, on which the maximum is reached in (1.4).

By Theorem 1, $R_2(x_0)$ is not empty. The set appearing in (2.4) is $T=\{[y, v] \mid y\in R_2(x_0),\ v\in V_2(y)\}$. Notice also that the following holds for $\Lambda_2(y)$:

$$\begin{aligned}\Lambda_2(y)=\Bigg\{\lambda \,\Bigg|\, &\sum_{i=1}^{m} \lambda_i h_{iy}(x_0, y)=f_y(x_0, y),\ \lambda_i\geqslant 0,\ \lambda_i h_i(x_0, y)=0,\\ &\lambda_i[(h_{iy}(x_0, y), v)+(h_{ix}(x_0, y), g)]=0, \quad i=1, 2, \ldots, m\Bigg\},\end{aligned} \tag{3.5}$$

where v is an arbitrary solution of the problem (3.1), (3.2), i.e., $v\in V_2(y)$.

Theorem 5

Under the conditions of Theorem 4,

$$\begin{aligned}\frac{\partial^2\varphi(x_0)}{\partial g^2} &= \max_{y\in R_2(x_0)} \left[\max_{v\in V_2(y)} \min_{\lambda\in\Lambda_2(y)} d^2L(y, v, \lambda) \right]\\ &= \max_{y\in R_2(x_0)} \left[\min_{\lambda\in\Lambda_2(y)} \max_{v\in V_2(y)} d^2L(y, v, \lambda) \right],\end{aligned} \tag{3.6}$$

where

$$\begin{aligned}d^2L(y, v, \lambda)&=(L_{yy}(x_0, y, \lambda)\ v, v)+2(L_{xy}(x_0, y, \lambda)v, g)\\ &+(L_{xx}(x_0, y, \lambda)g, g),\\ L(x, y, \lambda)&=f(x, y)-\sum_{i=1}^{m} \lambda_i h_i(x, y).\end{aligned}$$

Proof. Consider expression (2.4). The discovery of sup (f_y, w) with respect to $w\in\Gamma_2(y, v)$ is a problem of linear programming, by virtue of Lemma 3. The set of admissible solutions of the dual problem is the set (3.5). Hence, with $[y, v]\in T$

$$\max_{w\in\Gamma_2(y,v)} 2(f_y, w) = \min_{\lambda\in\Lambda_2(y)} \left[-\sum_{i=1} \lambda_i d^2 h_i \right].$$

The first equation in (3.6) is now proved. Let us show that the maximum and minimum in the brackets can be interchanged. Let $y\in R(x_0)$. The set $\Lambda(y)$, and hence the narrower set $\Lambda_2(y)$, are bounded. For, by (2.3), a v exists such that

$$(h_{iy}(x_0, y), v) \geqslant 1, \qquad i\in Q(y).$$

Hence, for any $\lambda\in\Lambda(y)$ we have

$$\sum_{i=1}^{m} \lambda_i \leqslant \sum_{i=1}^{m} \lambda_i (h_{iy}(x_0, y), v) = (f_y(x_0, y), v).$$

Hence $\Lambda_2(y)$ is a convex compactum, and the function $d^2L(y, v, \lambda)$ is concave with respect to v and linear with respect to λ. Hence

$$\sup_{v\in V_2(y)} \inf_{\lambda\in\Lambda_2(y)} d^2L(y, v, \lambda) = \inf_{\lambda\in\Lambda_2(y)} \sup_{v\in V_2(y)} d^2L(y, v, \lambda)$$

(see [2], p. 13). The suprema with respect to $v\in V_2(y)$ are reached, since $f_{yy}(x_0,y)$ is negative definite, while the infima with respect to $\lambda\in\Lambda_2(y)$ are reached because $\Lambda_2(y)$ is compact. The theorem is proved.

Note. If $\Omega(x)$ is independent of x:

$$\Omega(x) \equiv \Omega = \{y \mid h_i(y) \leqslant 0,\ i=1, 2, \ldots, m\},$$

then the sets R_2, V_2, and Λ_2 in Theorem 5 are specified as follows:

$$R_2(x_0) = \left\{ y\in R(x_0) \;\middle|\; \frac{\partial\varphi(x_0)}{\partial g} = (f_x(x_0, y), g) \right\},$$

$$V_2(y) = \{v \mid (h_{iy}(y), v) \leqslant 0,\ i\in Q(y),\ (f_y(x_0, y), v) = 0\},$$

$$\Lambda_2(y) = \Lambda(y) = \left\{ \lambda \;\middle|\; \sum_{i=1}^{m} \lambda_i h_{iy}(y) = f_y(x_0, y),\ \lambda_i \geqslant 0,\ \lambda_i h_i(y) = 0 \right\}$$

for arbitrary $y\in R(x_0)$. If, at the same time, all the functions $H_i(y)$ are linear d^2L can be replaced by d^2L can be replaced by d^2f and the minimum with respect to $\lambda\in\Lambda_2(y)$ disappears. The relevant expression was obtained in [8].

4. Higher-order derivatives

Let $n \geqslant 2$ and suppose that we have already obtained the expansion

$$\varphi(x_0+tg) = \varphi(x_0) + \sum_{i=1}^{n-1} \frac{\partial^i\varphi(x_0)}{\partial g^i} \frac{t^i}{i!} + o(t^{n-1}),$$

where $\partial^i\varphi(x_0)/\partial g^i$ are certain numbers. Then the limit

$$\frac{\partial^n\varphi(x_0)}{\partial g^n} = \lim_{t\to+0} \frac{n!}{t^n} \left[\varphi(x_0+tg) - \varphi(x_0) - \sum_{i=1}^{n-1} \frac{\partial^i\varphi(x_0)}{\partial g^i} \frac{t^i}{i!} \right]$$

if it exists, will be called the n-th directional derivative.

We introduce the set $\Gamma_n(y, v_1, \ldots, v_{n-1})$ of admissible n-th order directions. Suppose the sets $\Gamma_1, \ldots, \Gamma_{n-1}$, have already been determined. We fix $y\in\Omega(x_0)$, $v_1\in\Gamma_1(y), \ldots, v_{n-1}\in\Gamma_{n-1}$

$(y, v_1, \ldots, v_{n-2})$ and introduce the set

$$\gamma_n(y, v_1, \ldots, v_{n-1})$$
$$= \left\{ v_n \mid \exists t_0 > 0: \; y + \sum_{i=1}^{n} t^i v_i \in \Omega(x_0 + tg) \quad \text{for} \quad 0 \leqslant t < t_0 \right\}$$

We denote the closure of $\gamma_n(y, v_1, \ldots, v_{n-1})$ by $\Gamma_n(y, v_1, \ldots, v_{n-1})$ and call it the set of admissible n-th order directions.

Assume that the function f is n times continuously differentiable. Then, given any $\nu = 1, 2, \ldots, n$ we have the expansion

$$f\left(x_0 + tg, y + \sum_{i=1}^{\nu} t^i v_i\right) = f(x_0, y) + \sum_{i=1}^{\nu} A_i(y, v_1, \ldots, v_i) \frac{t^i}{i!} + o(t^\nu), \tag{4.1}$$

where the A_i are independent of ν and are expressible in terms of the derivatives of f of order not greater than i.

Let us also assume that, given any sequence $\{y_s\}$, $y_s \in R(x_0 + t_s g)$, $t_s \to +0$, a sequence $\{y_k\}$, exists representable in the form

$$y_k = \bar{y} + \sum_{i=1}^{n-1} t_k^i \bar{v}_i + t_k^n v_{nk} + o(t_k^n),$$

where $\bar{y} \in \Omega(x_0)$, $\bar{v}_1 \in \Gamma_1(\bar{y}), \ldots, \bar{v}_{n-1} \in \Gamma_{n-1}(\bar{y}, \bar{v}_1, \ldots, \bar{v}_{n-2})$, $v_{nk} \in \Gamma_n(\bar{y}, \bar{v}_1, \ldots, \bar{v}_{n-1})$, $t_k v_{nk} \to 0$ as $k \to \infty$.

Notice that the proof of Theorem 4 amounts in essence to verifying this condition for $n = 2$.

Theorem 6

If If $n \geqslant 2$, a first directional derivative exists, and (1.4) holds, then, under the above assumptions, derivatives of any order $\nu = 2, 3, \ldots, n$, will exist and we have

$$\frac{\partial^\nu \varphi(x_0)}{\partial g^\nu} = \sup_{[y, v_1, \ldots, v_{\nu-1}] \in T_{\nu-1}} \; \sup_{v_\nu \in \Gamma_\nu(y, v_1, \ldots, v_{\nu-1})} A_\nu(y, v_1, \ldots, v_\nu), \tag{4.2}$$

where the $A_\nu(y, v_1, \ldots, v_\nu)$ come from expression (4.1). $T_{\nu-1}$ is the set of collections $[y, v_1, \ldots, v_{\nu-1}]$, on which the supremum is reached in the expression for the $(\nu-1)$-th derivative, and $T_{\nu-1}$ is not empty for $\nu = 1, 2, \ldots, n$.

Equations (4.2) are proved successively: first for $\nu = 2$, then for $\nu = 3$, etc. The scheme of the proof is the same as in Theorem 4.

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THE LINEAR TIME OPTIMAL PROBLEM WITH A SMALL PARAMETER*

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THE STRUCTURE of the isochronous surfaces and the disturbance of the optimal controls in the linear time optimal problem are examined, when a small change occurs in the conditions.

Let γ be a small numerical parameter. Consider the following time optimal problem, to be denoted by $P(\gamma)$:

It is required to move from the point $x_0(\gamma)$ to the point 0 in minimum time under the constraints

$$\dot{x}=A(t,\gamma)x+B(t,\gamma)u, \tag{0.1}$$

where x and u are n-dimensional column vectors, and A and B are $n\times r$ matrices respectively.

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